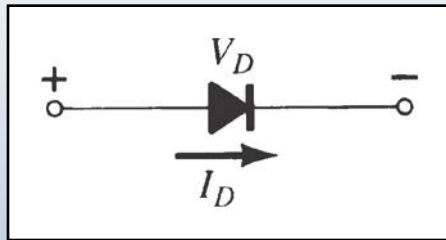


# **Semiconductor Diodes**

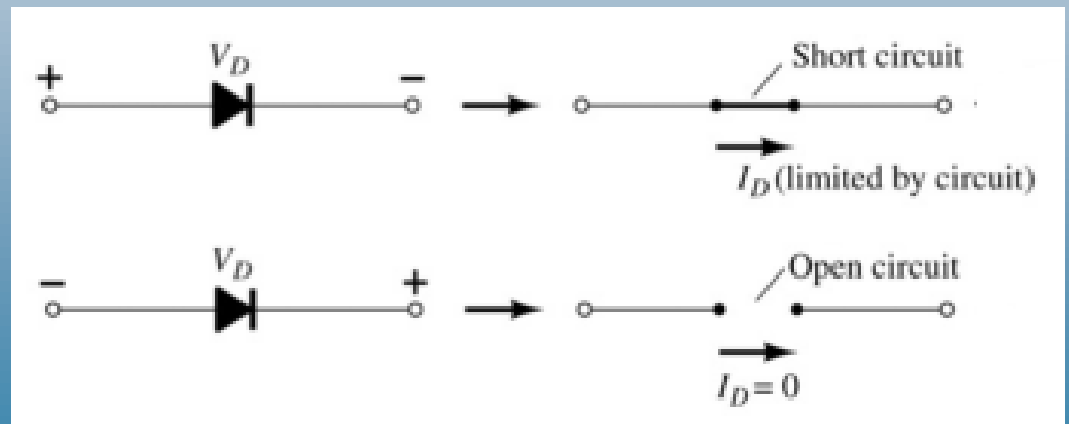
## **Topic 1 (Chapter 1)**

# Diodes

The diode is a 2-terminal device.

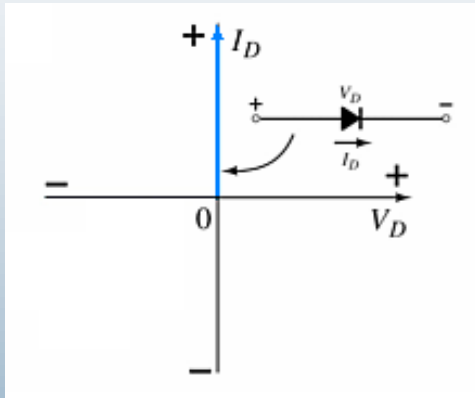


A diode ideally conducts in only one direction.



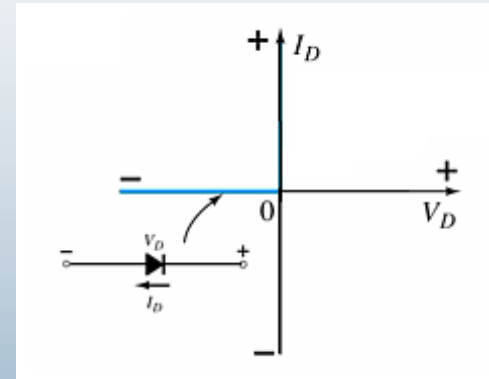
# Diode Characteristics

## Conduction Region



- The voltage across the diode is 0 V
- The diode acts like a short

## Non-Conduction Region



- All of the voltage is across the diode
- The current is 0 A
- The diode acts like open

# 1.2 Semiconductor Materials

Materials commonly used in the development of semiconductor devices:

Silicon (Si)

Germanium (Ge)

Gallium Arsenide (GaAs)

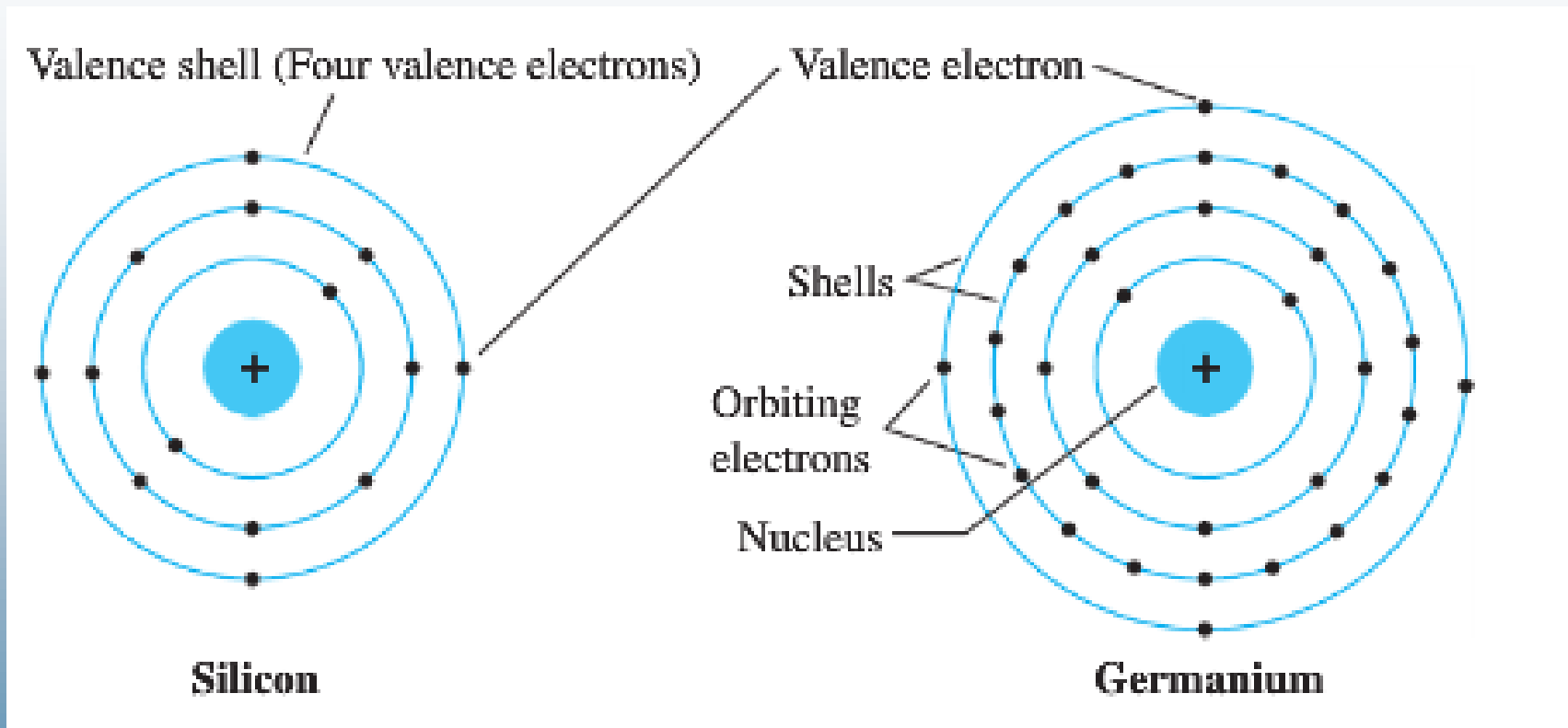
# What are Semiconductors?

- Semiconductors are a group of materials having electrical conductivities intermediate between metals (conductors) and insulators.
- Their conductivities can be varied by changes in:
  - temperature,
  - impurity content.

# History of Popular Semiconductors

- **Ge (Germanium)** was widely used in the early days.
- **Si (Silicon)** is now used for the majority of integrated circuits (ICs).
- The compound semiconductors are widely used in high-speed devices and opto-electronic devices
  - For example, **III–V semiconductors** such as GaN, GaP, and GaAs are common in light- emitting diodes (LEDs)
- **Three-element or ternary** semiconductors (such as GaAsP) and **four-element or quaternary** semiconductors (such as InGaAsP) are also used.
  - For example, they can be used to make LEDs of different colors.

# 1.3 Atomic Structures of Semiconductors

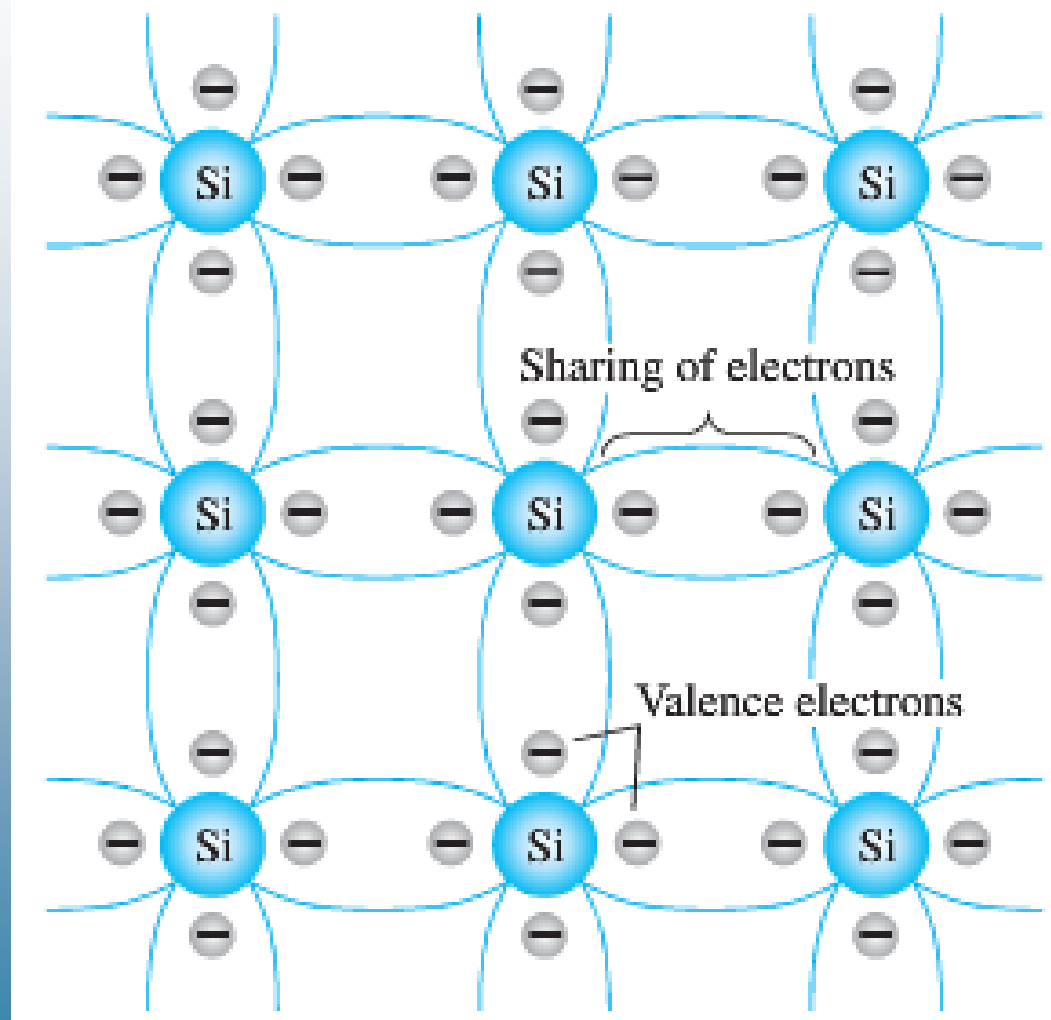


**Valence electrons: Electrons in the outermost shell of an atom.**  
**Si and Ge are Tetravalent atoms with four valence electrons**

$$Z_{Si} = 14$$

$$Z_{Ge} = 32$$

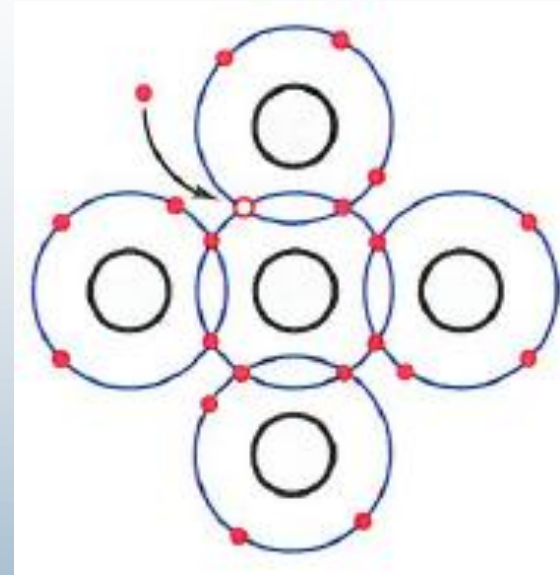
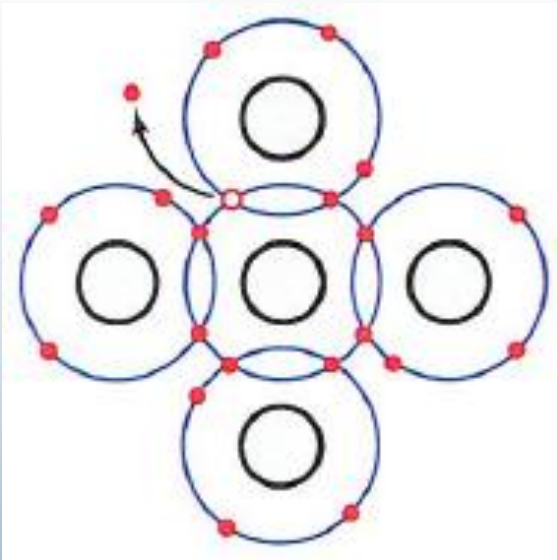
# Covalent bonding of the silicon atom



# Intrinsic Semiconductors

- Semiconductor materials two types:
  1. Intrinsic Semiconductors
  2. Extrinsic Semiconductors
- A **pure** semiconductor
- Carefully refined to reduce the number of impurities to a very low level
- Intrinsic carriers: free electrons in a material due only to external causes (energy from light or heat)

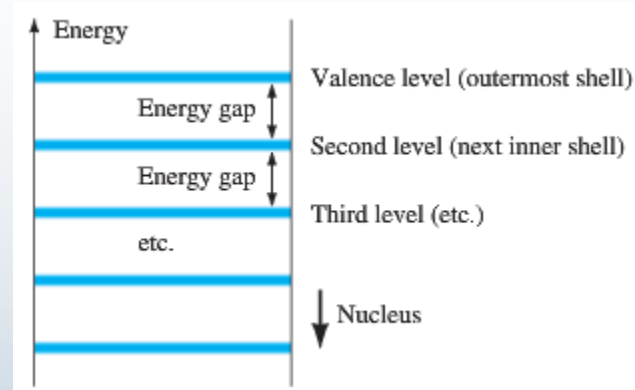
# Electron Hole Pair Generation and Recombination



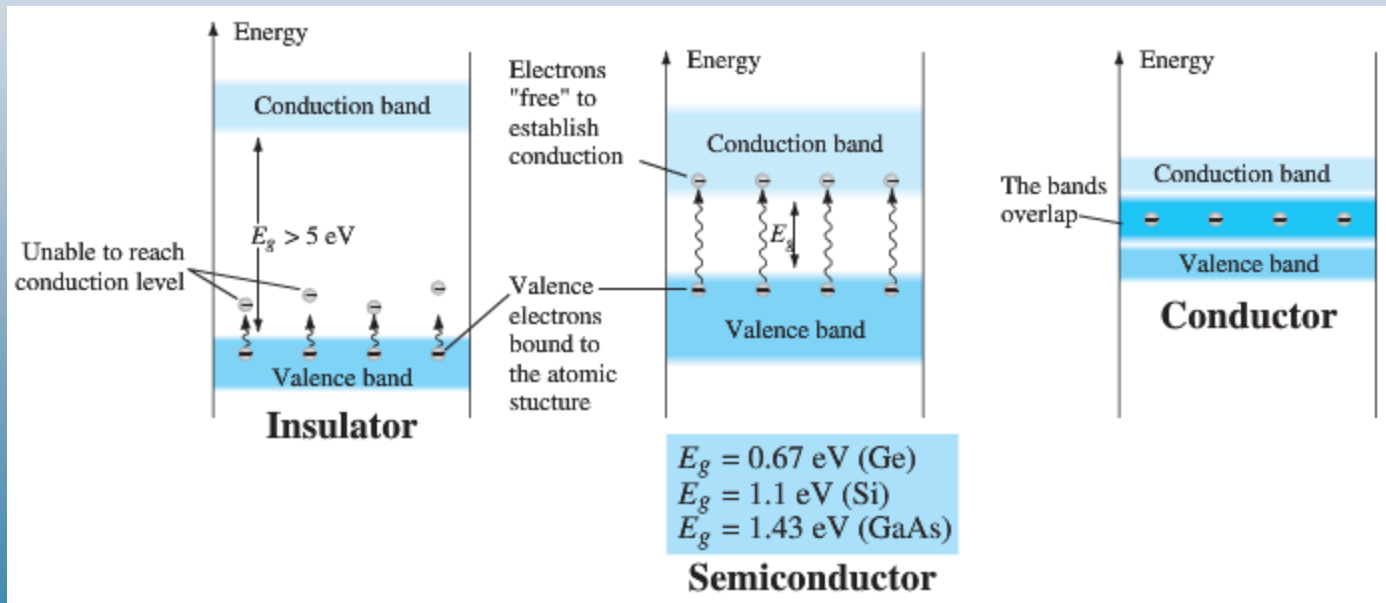
- Valence electrons can absorb sufficient energy (from light or heat) to break the covalent bonds and assume the “free” state.
  - These free electrons are called **intrinsic carriers**
- Higher temperatures creates more intrinsic carriers – hence higher conductivity

# 1.4 ENERGY LEVELS

- The farther an electron is from the nucleus, the higher is the energy state



(a) Discrete energy levels in isolated atomic structures



(b) conduction and valence bands of an insulator, a semiconductor, and a conductor.

# 1.5 n-TYPE AND p-TYPE MATERIALS

- The electrical properties of a semiconductor can be altered significantly by adding impurity atoms
  - This process is called **doping**
  - A **doped** semiconductor is called **extrinsic**
- There are just two types of extrinsic semiconductor materials:

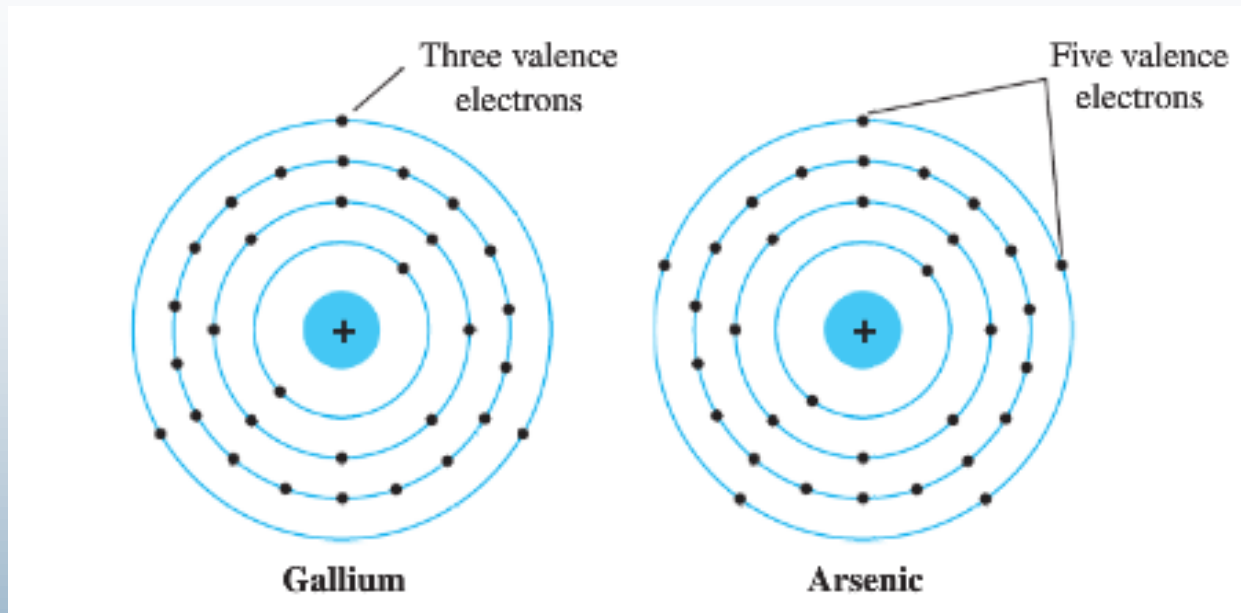
***n*-type**

***n*-type** materials contain an excess of conduction band **electrons**.

***p*-type**

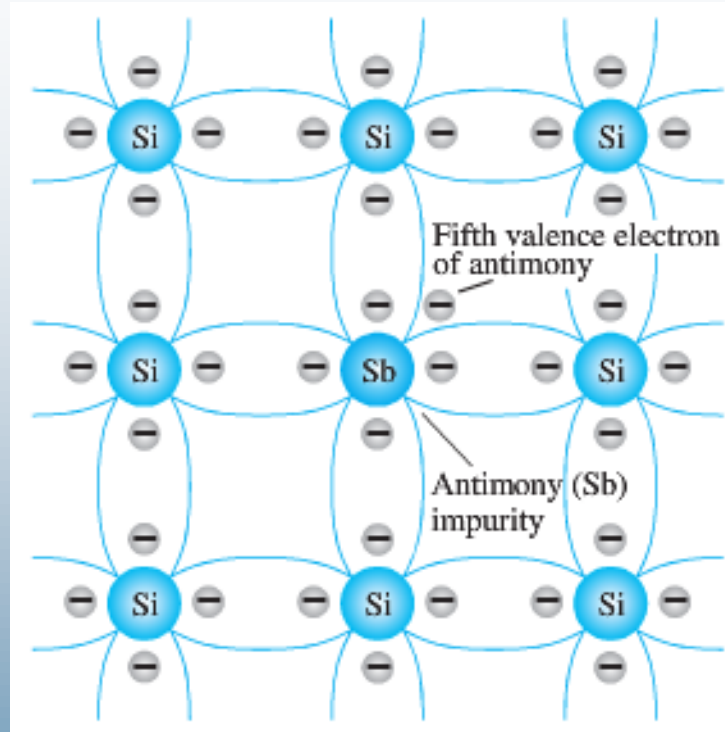
***p*-type** materials contain an excess of valence band **holes**.

# Atomic Structures of Impurities



- p-type material is created by adding trivalent impurity elements such as B, Ga, and In
- n-type material is created by adding pentavalent impurity elements such as Sb, As, and P

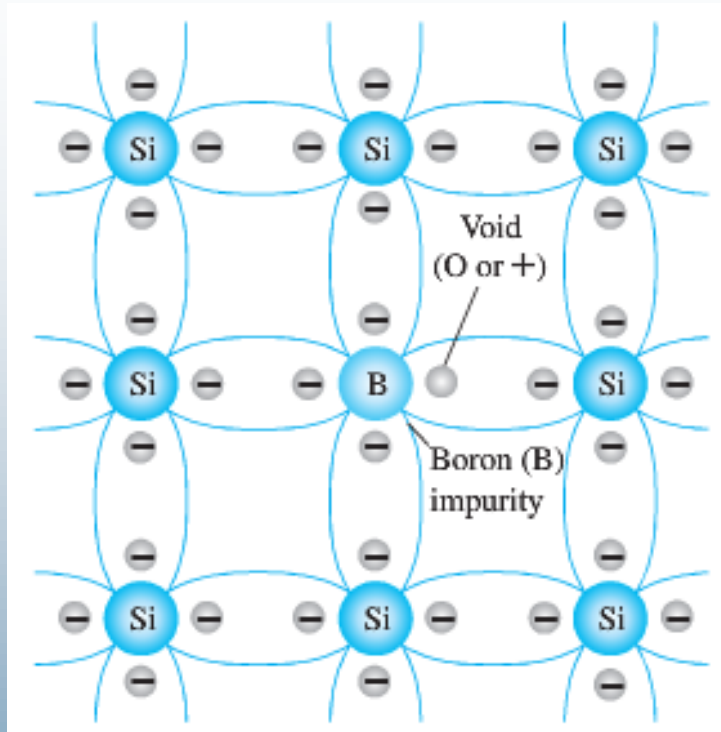
# n -Type Material



Antimony impurity in n-type material.

- Impurities with five valence electrons (pentavalent) are called **donor** atoms, because the inserted impurity atom has donated a relatively “free” electron to the structure

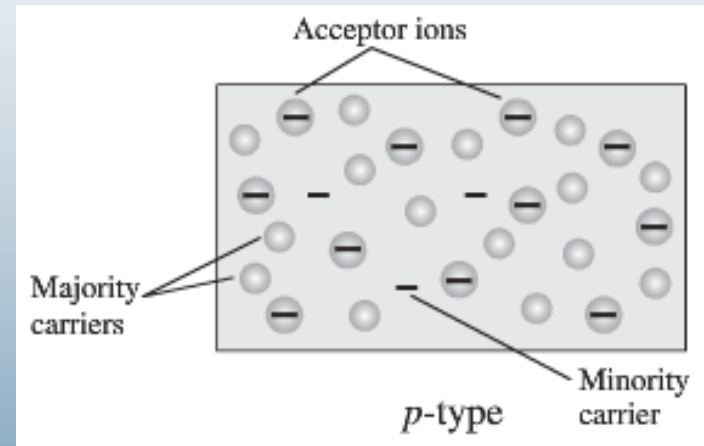
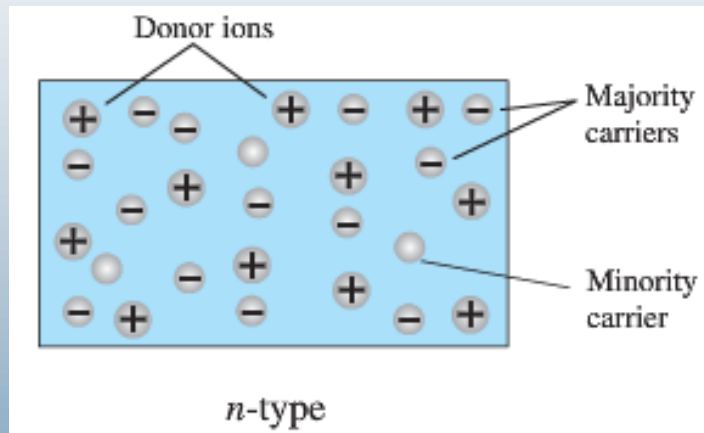
# p-Type Material



Boron impurity in p-type material.

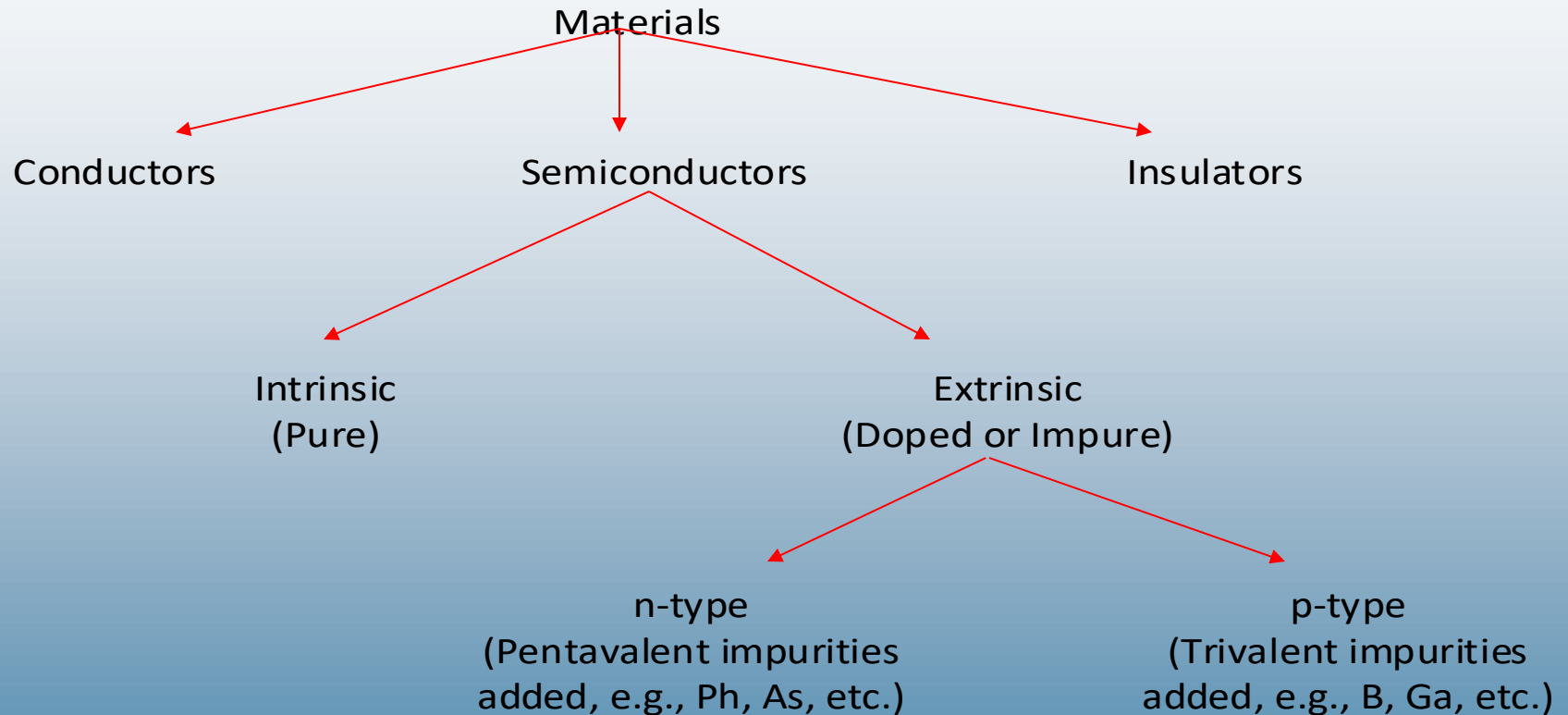
- There is now an insufficient number of electrons to complete the covalent bonds of the newly formed lattice.
  - The resulting vacancy is called a **hole**
- The diffused impurities with three valence electrons are called **acceptor** atoms, because the resulting vacancy will readily accept a free electron

# Majority and Minority Carriers



- In an **n-type** material, the **electron** is called the **majority** carrier and the hole the minority carrier.
- In a **p-type** material, the **hole** is called the **majority** carrier and the electron the minority carrier

# Electrical Classification of Materials



# Semiconductors in Summary

- The most popular material is **silicon**.
- **Pure** crystals are intrinsic semiconductors.
- **Doped** crystals are extrinsic semiconductors.
- Crystals are doped to be **n** type or **p** type.
- A doped semiconductor will have mostly majority carriers and a few thermally generated *minority* carriers.
  - **Electrons** are majority carriers in **n** type
  - **Holes** are majority carriers in **p** type

## 2. PROPERTIES OF SILICON AND GERMANIUM AT 300 K

| Property                                                          | Silicon (Si)         | Germanium (Ge)       |
|-------------------------------------------------------------------|----------------------|----------------------|
| Energy Gap, $E_g$ (eV)                                            | 1.1                  | 0.7                  |
| Electron Mobility, $\mu_n$ ( $\text{m}^2/\text{V}\cdot\text{s}$ ) | 0.135                | 0.39                 |
| Hole Mobility, $\mu_p$ ( $\text{m}^2/\text{V}\cdot\text{s}$ )     | 0.048                | 0.19                 |
| Intrinsic Carrier Density, $n_i$ ( $\text{m}^{-3}$ )              | $1.5 \times 10^{16}$ | $2.4 \times 10^{19}$ |
| Intrinsic Resistivity ( $\Omega\cdot\text{m}$ )                   | 2300                 | 0.46                 |
| Atomic Density ( $\text{atoms}/\text{m}^3$ )                      | $2.3 \times 10^{28}$ | $4.4 \times 10^{28}$ |

### OBSERVATIONS

- ① Germanium has a smaller bandgap than silicon.
- ② Germanium possesses higher carrier mobility.
- ③ Germanium has a much larger intrinsic carrier concentration.
- ④ Silicon exhibits much higher intrinsic resistivity.
- ⑤ Silicon is preferred in electronics because of better thermal stability, lower leakage current and easy oxide formation ( $\text{SiO}_2$ ).

### 3. INTRINSIC SEMICONDUCTOR

An intrinsic semiconductor is a pure semiconductor without any impurity atoms.

At thermal equilibrium:

$$n = p = n_i$$

$n$  = electron concentration

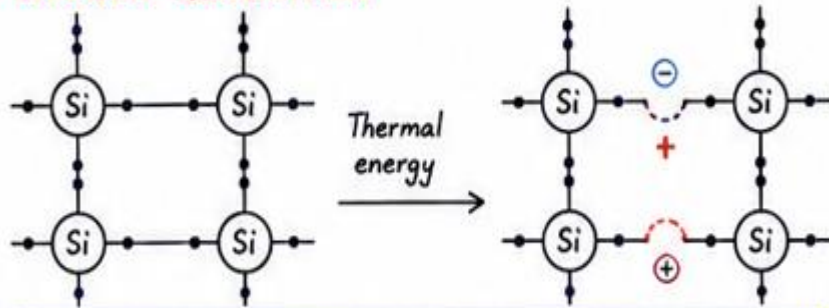
$p$  = hole concentration

$n_i$  = intrinsic carrier concentration

Thus:

$$n = p = n_i$$

#### CARRIER GENERATION



- Thermal energy breaks some covalent bonds.
- One free electron (-) is generated.
- One hole (+) is generated.

Therefore, number of electrons = number of holes.

### 4. MASS ACTION LAW

#### STATEMENT

Under thermal equilibrium, the product of free electron concentration ( $n$ ) and free hole concentration ( $p$ ) remains constant and equals the square of the intrinsic carrier concentration.

$$np = n_i^2$$

(Mass Action Law)

#### SIGNIFICANCE

Even when doping changes electron and hole concentrations individually:

- If  $n$  increases,  $p$  decreases.
- If  $p$  increases,  $n$  decreases.

But, their product  $np$  remains constant and equal to  $n_i^2$  at thermal equilibrium.

$$np = n_i^2 \text{ (always true at thermal equilibrium)}$$

## 5. CHARGE NEUTRALITY CONDITION

A semiconductor remains electrically neutral.

$$\text{Total positive charge} = \text{Total negative charge}$$

For a doped semiconductor:

$$N_D + p = N_A + n$$

Where

$N_D$  = donor concentration (positive ionized donors)

$N_A$  = acceptor concentration (negative ionized acceptors)

$n$  = electron concentration

$p$  = hole concentration

## 7. CARRIER CONCENTRATION IN N-TYPE SEMICONDUCTOR

For n-type material:

$$N_A = 0$$

Charge neutrality:

$$N_D + p = n$$

Since  $n \gg p$  (electrons are majority)

$$n \approx N_D$$

Mass Action Law:

$$np = n_i^2$$

Substitute  $n \approx N_D$ :

$$N_D p = n_i^2$$

$$p_n = \frac{n_i^2}{N_D}$$

### SUMMARY (n-type)

Majority carriers : Electrons

Minority carriers : Holes

$$n_n \approx N_D$$

$$p_n = \frac{n_i^2}{N_D}$$

$N_D$  = donor concentration

$n_n$  = majority electron concentration

$p_n$  = minority hole concentration

## 8. CARRIER CONCENTRATION IN P-TYPE SEMICONDUCTOR

For p-type material:

$$N_D = 0$$

Charge neutrality:

$$N_D + p = N_A + n$$

$$\Rightarrow p = N_A + n$$

Since  $p \gg n$  (holes are majority)

$$p \approx N_A$$

Mass Action Law:

$$np = n_i^2$$

Substitute  $p \approx N_A$ :

$$n N_A = n_i^2$$

$$n_p = \frac{n_i^2}{N_A}$$

### SUMMARY (p-type)

Majority carriers : Holes

Minority carriers : Electrons

$$p_p \approx N_A$$

$$n_p = \frac{n_i^2}{N_A}$$

$N_A$  = acceptor concentration

$p_p$  = majority hole concentration

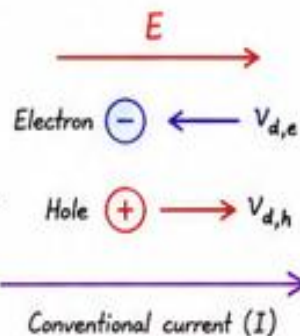
$n_p$  = minority electron concentration

# Mobility, Drift Velocity and Current Density

## Mobility ( $\mu$ )

When an electric field ( $E$ ) is applied:

- Electrons move opposite to the field.
- Holes move in the direction of the field.
- Both produce current in the same direction.



## Drift velocity ( $v_d$ )

$$v_d \propto E \quad \text{or} \quad v_d = \mu E$$

where  $v_d$  = drift velocity (m/s)  
 $E$  = electric field (V/m)  
 $\mu$  = mobility ( $\text{m}^2/\text{V}\cdot\text{s}$ )

## Mobility

$$\mu = \frac{v_d}{E} \quad \text{Unit: } \text{m}^2/\text{V}\cdot\text{s}$$

Typically

$$\mu_n > \mu_p$$

(electrons are more mobile than holes)

## Current Density

### Electron current density

$$\begin{aligned} J_n &= nq v_{d,e} \\ &= nq (\mu_n E) \end{aligned}$$

$$J_n = nq \mu_n E$$

### Hole current density

$$\begin{aligned} J_p &= pq v_{d,h} \\ &= pq (\mu_p E) \end{aligned}$$

$$J_p = pq \mu_p E$$

### Total current density

$$J = J_n + J_p = (nq \mu_n + pq \mu_p) E$$

$$J = \sigma E$$

where

$$\sigma = q(n\mu_n + p\mu_p) \quad (\text{conductivity})$$

Unit of  $J$ :  $\text{A}/\text{m}^2$ , Unit of  $\sigma$ :  $(\Omega\cdot\text{m})^{-1}$

## KEY POINTS

- High conductivity  $\Rightarrow$  Low resistivity
- Low conductivity  $\Rightarrow$  High resistivity
- Intrinsic semiconductors have high resistivity.
- Doping increases conductivity and decreases resistivity.

$$\rho = \frac{1}{\sigma} \quad \text{— Resistivity is the reciprocal of conductivity.}$$

### Example 1: Intrinsic Conductivity of Germanium

Question :

[5 Marks]

The electron mobility and hole mobility of an intrinsic germanium semiconductor at room temperature are  $0.38 \text{ m}^2/\text{V}\cdot\text{s}$  and  $0.18 \text{ m}^2/\text{V}\cdot\text{s}$ , respectively.

If the intrinsic carrier concentration is  $2.5 \times 10^{19} \text{ m}^{-3}$ , calculate the intrinsic conductivity of germanium.

Take the electronic charge as  $q = 1.6 \times 10^{-19} \text{ C}$ .

Solution :

For an intrinsic semiconductor,

$$\sigma_i = q n_i (\mu_n + \mu_p)$$

Given :

$$\mu_n = 0.38 \text{ m}^2/\text{V}\cdot\text{s}$$

$$\mu_p = 0.18 \text{ m}^2/\text{V}\cdot\text{s}$$

$$n_i = 2.5 \times 10^{19} \text{ m}^{-3}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$\sigma_i$  = intrinsic conductivity (S/m)

$q$  = electronic charge (C)

$n_i$  = intrinsic carrier concentration ( $\text{m}^{-3}$ )

$\mu_n$  = electron mobility ( $\text{m}^2/\text{V}\cdot\text{s}$ )

$\mu_p$  = hole mobility ( $\text{m}^2/\text{V}\cdot\text{s}$ )

Substituting the values,

$$\begin{aligned}\sigma_i &= (1.6 \times 10^{-19}) (2.5 \times 10^{19}) (0.38 + 0.18) \\ &= (1.6 \times 2.5) (0.56) \\ &= 4 \times 0.56 \\ &= 2.24\end{aligned}$$

Therefore,  $\sigma_i = 2.24 (\Omega\cdot\text{m})^{-1}$   
or  
 $\sigma_i = 2.24 \text{ S/m}$

Final Answer :

$$\sigma_i = 2.24 \text{ S/m}$$

[CO-1 | 5 Marks]

## Example 2: Donor-Doped Germanium Semiconductor

[10 Marks]

A germanium semiconductor sample is doped with donor impurities at the rate of 1 donor atom per  $10^8$  germanium atoms.

The following data are given:

$$\begin{aligned}\mu_n &= 3800 \text{ cm}^2/\text{V}\cdot\text{s} & N_{\text{Ge}} &= 4.41 \times 10^{22} \text{ cm}^{-3} \\ \mu_p &= 1800 \text{ cm}^2/\text{V}\cdot\text{s} & q &= 1.6 \times 10^{-19} \text{ C} \\ n_i &= 2.5 \times 10^{13} \text{ cm}^{-3}\end{aligned}$$

Assuming complete ionization of donor atoms:

- Determine the donor concentration  $N_D$ . [2 Marks]
- Calculate the majority carrier concentration  $n$ . [2 Marks]
- Using the Mass Action Law, determine the minority carrier concentration  $p$ . [2 Marks]
- Calculate the conductivity  $\sigma$  of the semiconductor. [2 Marks]
- Determine the resistivity  $\rho$  of the semiconductor. [2 Marks]

### Solution:

#### (a) Donor Concentration

Since there is one donor atom for every  $10^8$  germanium atoms,

$$\begin{aligned}N_D &= \frac{N_{\text{Ge}}}{10^8} \\ &= \frac{4.41 \times 10^{22}}{10^8}\end{aligned}$$

$$N_D = 4.41 \times 10^{14} \text{ cm}^{-3}$$

[2 Marks]

#### (b) Majority Carrier Concentration

For an n-type semiconductor,

$$n \approx N_D$$

Therefore,

$$n = 4.41 \times 10^{14} \text{ cm}^{-3}$$

[2 Marks]

#### (c) Minority Carrier Concentration

Using Mass Action Law,

$$np = n_i^2$$

$$\begin{aligned}p &= \frac{n_i^2}{N_D} \\ &= \frac{(2.5 \times 10^{13})^2}{4.41 \times 10^{14}} \\ &= 1.42 \times 10^{12} \text{ cm}^{-3}\end{aligned}$$

$$p = 1.42 \times 10^{12} \text{ cm}^{-3}$$

[2 Marks]

#### (d) Conductivity

The conductivity of a semiconductor is

$$\sigma = q(n\mu_n + p\mu_p)$$

Substituting values,

$$\begin{aligned}\sigma &= (1.6 \times 10^{-19}) [(4.41 \times 10^{14})(3800) + \\ &\quad (1.42 \times 10^{12})(1800)] \\ &= (1.6 \times 10^{-19}) [1.6758 \times 10^{18} + 2.556 \times 10^{15}] \\ &= (1.6 \times 10^{-19}) (1.678356 \times 10^{18}) \\ &= 0.2685 (\Omega \cdot \text{cm})^{-1}\end{aligned}$$

$$\sigma \approx 0.27 (\Omega \cdot \text{cm})^{-1}$$

[2 Marks]

#### (e) Resistivity

$$\begin{aligned}\rho &= \frac{1}{\sigma} = \frac{1}{0.27} \\ &= 3.7037 \Omega \cdot \text{cm}\end{aligned}$$

$$\rho \approx 3.7 \Omega \cdot \text{cm}$$

#### Final Answer (Summary)

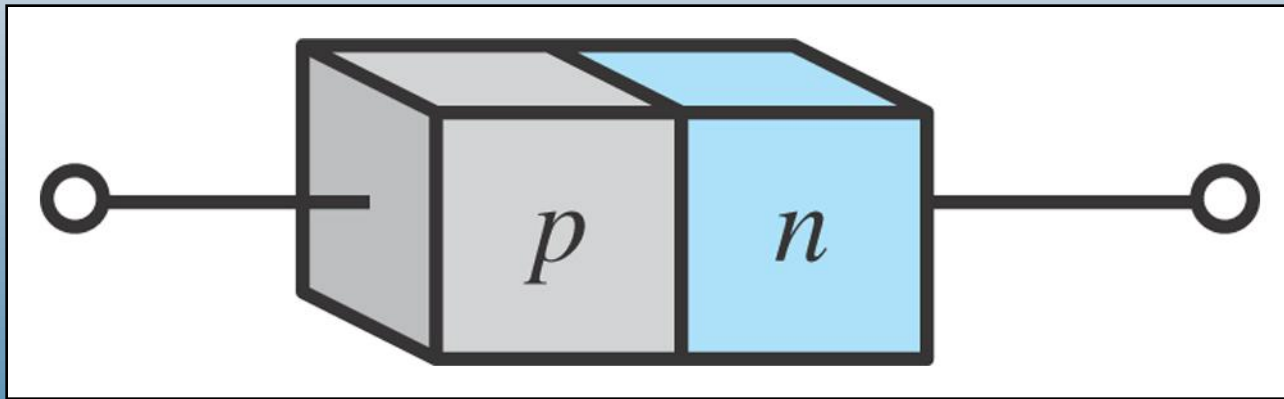
$$\begin{aligned}N_D &= 4.41 \times 10^{14} \text{ cm}^{-3} \\ n &= 4.41 \times 10^{14} \text{ cm}^{-3} \\ p &= 1.42 \times 10^{12} \text{ cm}^{-3} \\ \sigma &\approx 0.27 (\Omega \cdot \text{cm})^{-1} \\ \rho &\approx 3.7 \Omega \cdot \text{cm}\end{aligned}$$

[CO-1 | 10 Marks]

Formula Used:  $np = n_i^2$   $\sigma = q(n\mu_n + p\mu_p)$   $\rho = \frac{1}{\sigma}$

## 1.6 $p$ - $n$ Junctions

One end of a silicon or germanium crystal can be doped as a  $p$ -type material and the other end as an  $n$ -type material.



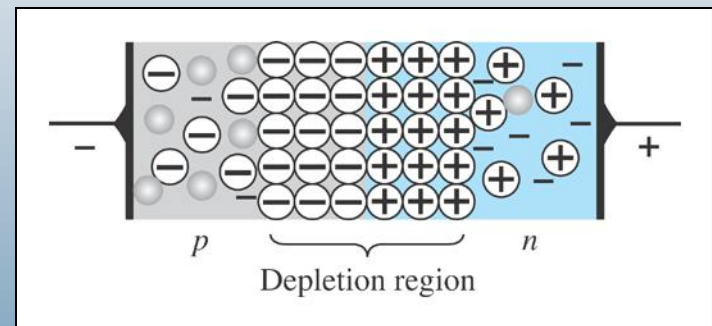
The result is a  $p$ - $n$  junction or Semiconductor Diode

# *p-n* Junctions

At the *p-n* junction, the excess conduction-band electrons on the *n*-type side are attracted to the valence-band holes on the *p*-type side.

The electrons in the *n*-type material migrate across the junction to the *p*-type material (electron flow).

Electron migration results in a **negative** charge on the *p*-type side of the junction and a **positive** charge on the *n*-type side of the junction.



**The result is the formation of a depletion region around the junction.**

# Diode Operating Conditions

A diode has three operating conditions:

1. No bias  $V_D = 0V$
2. Reverse bias  $V_D < 0V$
3. Forward bias  $V_D > 0V$

**Bias:** application of an external voltage ( $V_D$ )  
across the two terminals of the device

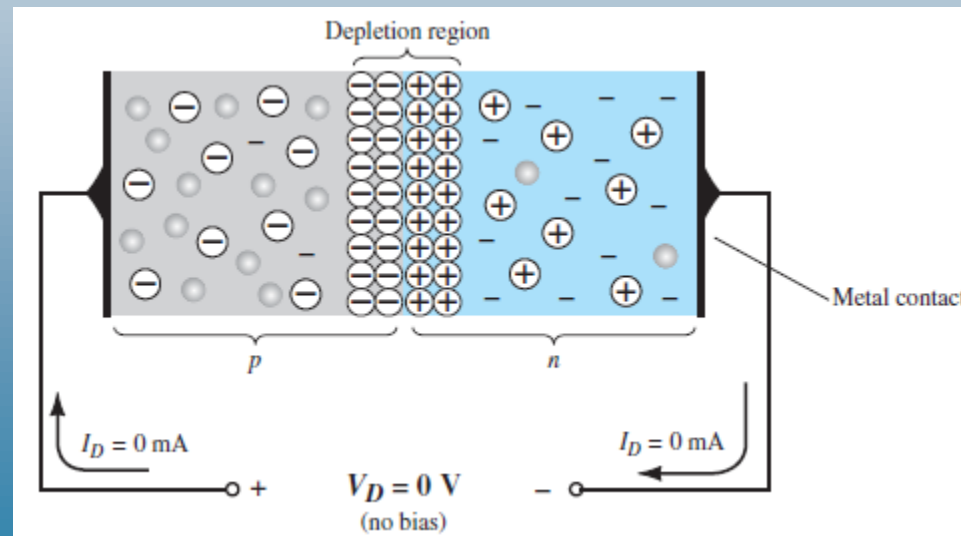
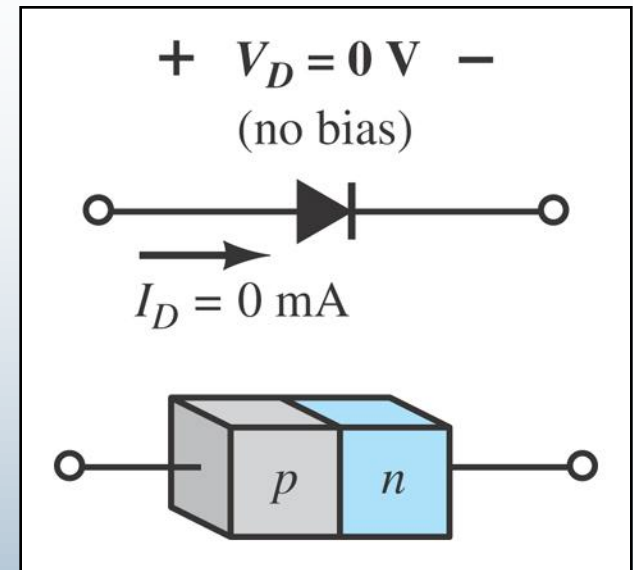
# Diode Operating Conditions

## No Bias

No external voltage is applied:  $V_D = 0 \text{ V}$

There is no diode current:  $I_D = 0 \text{ A}$

Only a modest depletion region exists

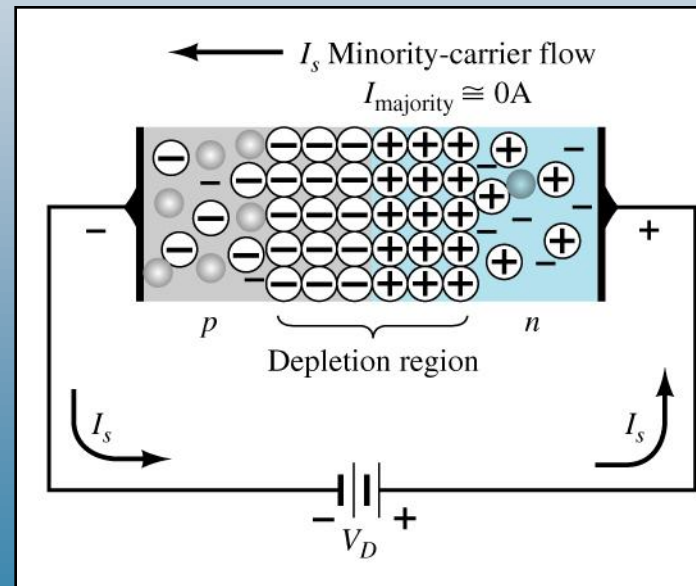
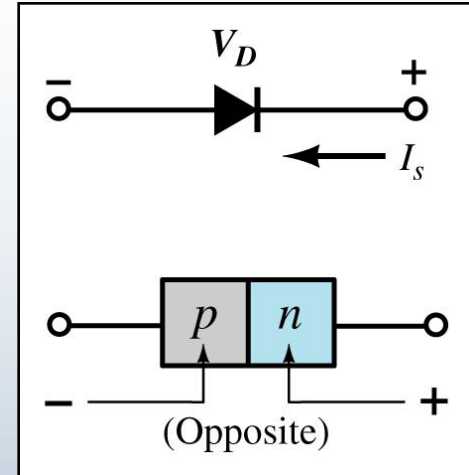


# Diode Operating Conditions

## Reverse Bias

External voltage is applied across the  $p$ - $n$  junction in the opposite polarity of the  $p$ - and  $n$ -type materials.

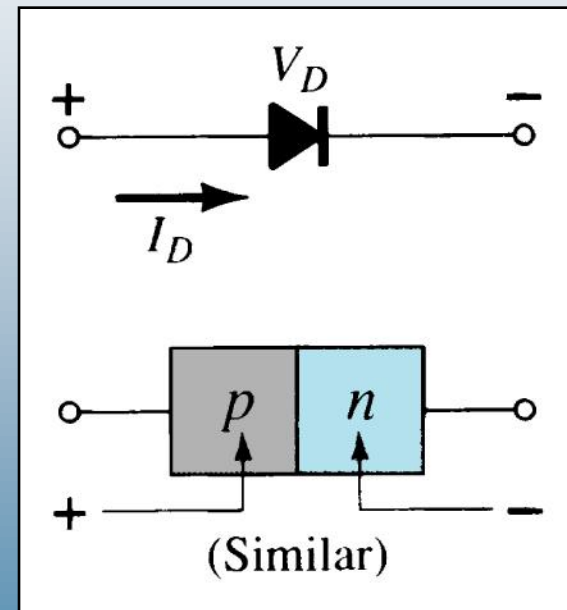
- The reverse voltage causes the depletion region to widen.
- $V_D < 0V$
- $I_{majority} = 0A$
- $I_D \cong I_{minority} = \text{small current}$



# Diode Operating Conditions

## Forward Bias

External voltage is applied across the  $p$ - $n$  junction in the same polarity as the  $p$ - and  $n$ -type materials.



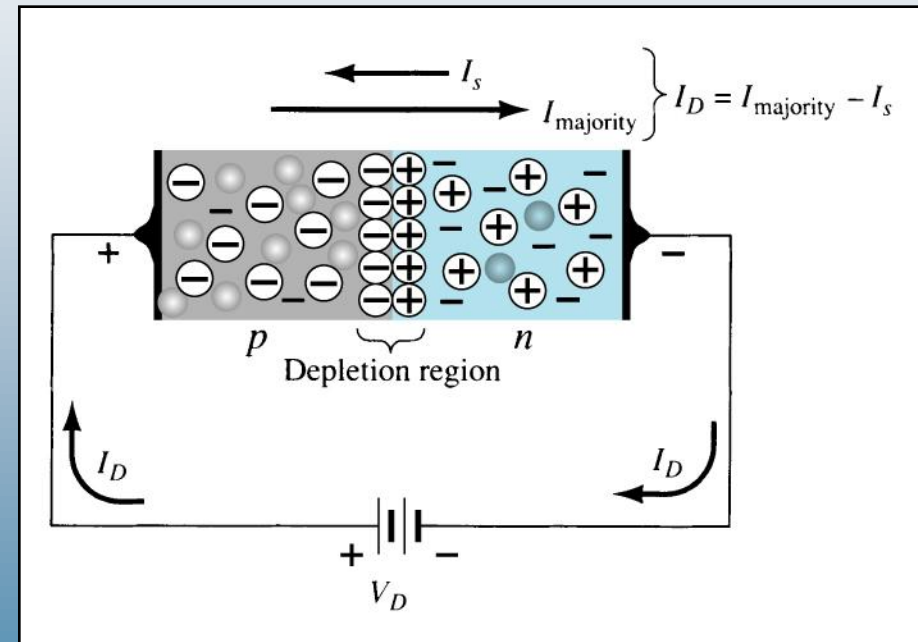
# Diode Operating Conditions

## Forward Bias

The forward voltage causes the depletion region to narrow.

- $V_D > 0V$
- $I_{majority} > 0A$
- $I_D \cong I_{majority} = \text{large current}$

The electrons and holes are pushed toward the  $p$ - $n$  junction.



The electrons and holes have sufficient energy to cross the  $p$ - $n$  junction.

# Diode Current Equation

It can be demonstrated through the use of solid-state physics that the general characteristics of a semiconductor diode can be defined by the following equation, referred to as Shockley's equation, for the forward- and reverse-bias regions:

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (\text{A}) \quad (1.2)$$

where  $I_s$  is the reverse saturation current  
 $V_D$  is the applied forward-bias voltage across the diode  
 $n$  is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ( $n = 1$  will be assumed throughout this text unless otherwise noted).

The voltage  $V_T$  in Eq. (1.1) is called the *thermal voltage* and is determined by

$$V_T = \frac{kT_K}{q} \quad (\text{V}) \quad (1.3)$$

where  $k$  is Boltzmann's constant =  $1.38 \times 10^{-23}$  J/K  
 $T_K$  is the absolute temperature in kelvins =  $273 +$  the temperature in  $^{\circ}\text{C}$   
 $q$  is the magnitude of electronic charge =  $1.6 \times 10^{-19}$  C

**EXAMPLE 1.1** At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage  $V_T$ .

**Solution:** Substituting into Eq. (1.3), we obtain

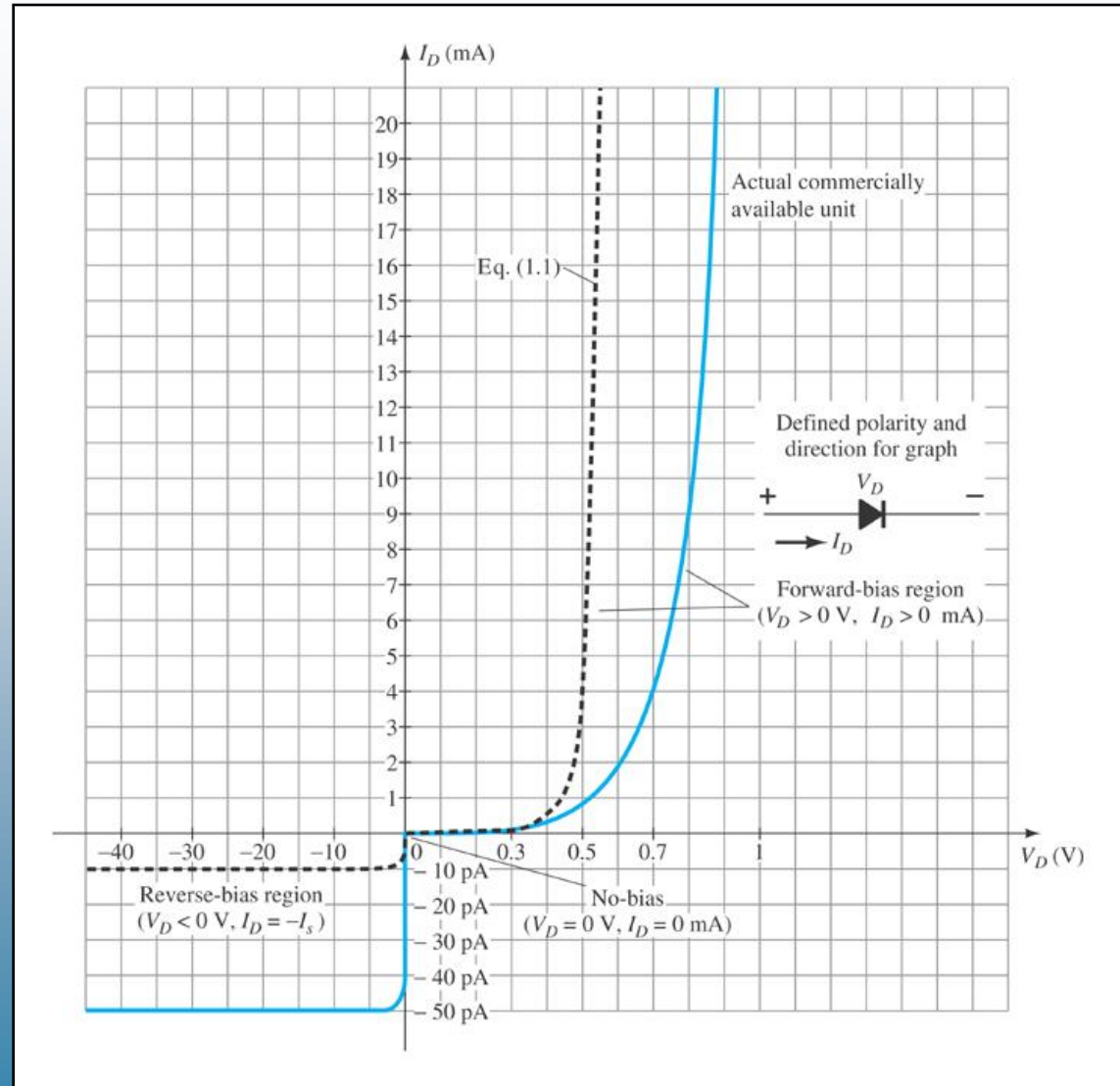
$$\begin{aligned} T &= 273 + ^\circ\text{C} = 273 + 27 = 300 \text{ K} \\ V_T &= \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \\ &= 25.875 \text{ mV} \cong 26 \text{ mV} \end{aligned}$$

The thermal voltage will become an important parameter in the analysis to follow in this chapter and a number of those to follow.

# Actual/Practical Diode Characteristics

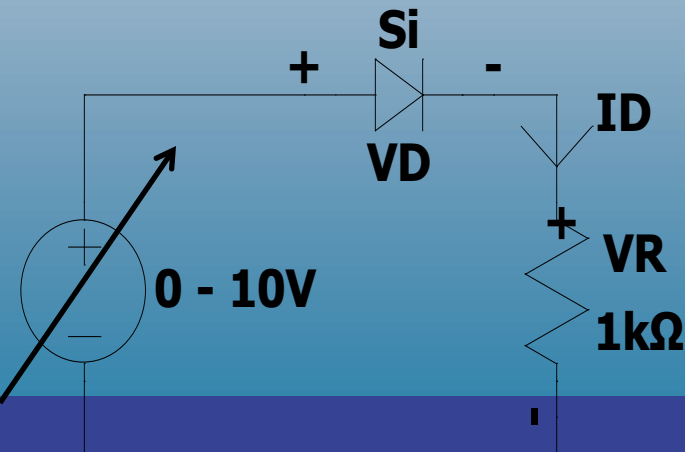
Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.



# Forward Bias (FB)/Short/Conduction Mode

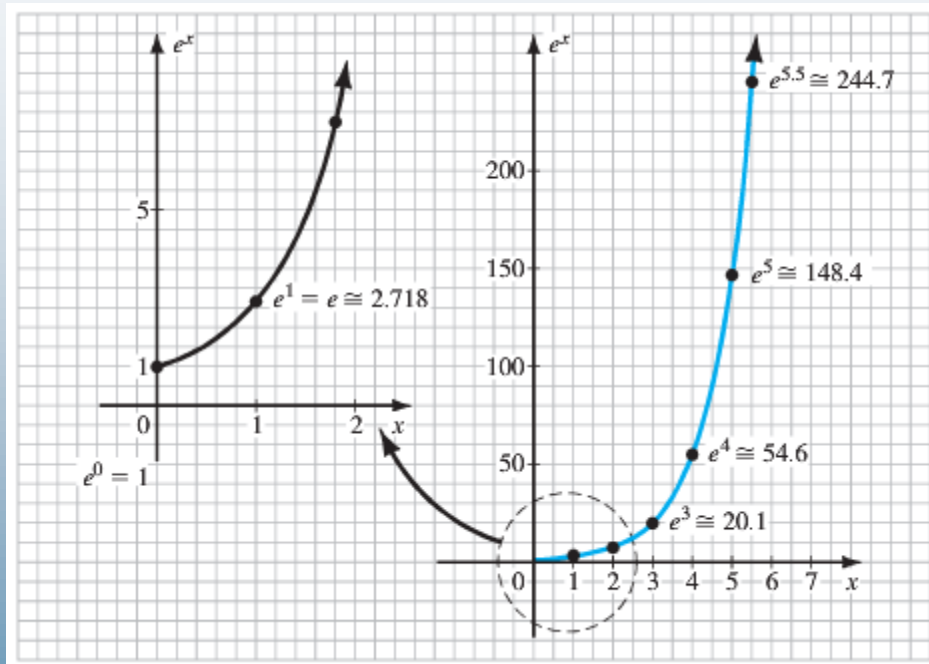
- $V_S < 0.7V$ : Diode  $\frac{NB}{OFF}$ .  $V_D = V_S$ .  $I_D = 0mA$ ;  $V_R = I_D R = 0V$
- $V_S \geq 0.7V$ : Diode: FB.  $V_D = 0.7V$ .
- $V_R = V_S - 0.7V$  (KVL)
- $I_D \neq 0mA = \frac{V_R}{R}$ ;



| $V_S$ (V) | $I_D$ (mA) | $V_D$ (V) | $V_R$ (V) | Diode State |
|-----------|------------|-----------|-----------|-------------|
| 0         | 0          | 0         | 0         | NB          |
| 0.1       | 0          | 0.1       | 0         | NB          |
| 0.3       | 0          | 0.3       | 0         | NB          |
| 0.5       | 0          | 0.5       | 0         | NB          |
| 0.71      | 0.01       | 0.7       | 0.01      | FB          |
| 1         | 0.3        | 0.7       | 0.3       | FB          |
| 5         | 4.3        | 0.7       | 4.3       | FB          |
| 10        | 9.3        | 0.7       | 9.3       | FB          |

# Diode Current Approximations

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (\text{A})$$



$$I_D = I_s e^{V_D/nV_T} - I_s$$

$$I_D \cong I_s e^{V_D/nV_T} \quad (V_D \text{ positive})$$

$$I_D \cong -I_s \quad (V_D \text{ negative})$$

| $V_D$                              | $e^{\frac{V_D}{nV_T}}$ | Value |
|------------------------------------|------------------------|-------|
| $-25 \text{ mV}$                   | $e^{-1}$               | 0.37  |
| $-50 \text{ mV}$                   | $e^{-2}$               | 0.14  |
| $-75 \text{ mV}$                   | $e^{-3}$               | 0.05  |
| $-100 \text{ mV} = -0.1 \text{ V}$ | $e^{-4}$               | 0.02  |

# Forward bias condition

- $I_D = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) = I_S e^{\frac{V_D}{nV_T}} - I_S \cong I_S e^{\frac{V_D}{nV_T}}$ ;  $I_S$  small current
- $I_1 = I_S e^{\frac{V_1}{nV_T}}$ ;  $I_1$  is the diode current for voltage  $V_1$
- $I_2 = I_S e^{\frac{V_2}{nV_T}}$ ;  $I_2$  is the diode current for voltage  $V_2$
- $\Delta V = V_2 - V_1 = nV_T \ln \frac{I_2}{I_1}$
- $\Delta V = V_2 - V_1 = 2.3 nV_T \log \frac{I_2}{I_1}$

# Forward bias condition

- Find the change in diode voltage if the current changes from  $0.1\mu A$  to  $10mA$  at room temperature with  $n = 1$ .
- $I_1 = 0.1\mu A = 0.1 \times 10^{-6} A$
- $I_2 = 10mA = 10 \times 10^{-3} A$
- $V_T = 26mV$  at room temperature
- $\Delta V = V_2 - V_1 = nV_T \ln \frac{I_2}{I_1} = 2.3 nV_T \log \frac{I_2}{I_1}$
- $\Delta V = 1 \times 26 \times 10^{-3} \ln \frac{10 \times 10^{-3}}{0.1 \times 10^{-6}} \neq 1 \times 26 \times 10^{-3} \ln \frac{10}{0.1}$

# Majority and Minority Carriers

*Two currents through a diode:*

## Majority Carriers

The majority carriers in **n-type** materials are electrons.

The majority carriers in **p-type** materials are holes.

## Minority Carriers

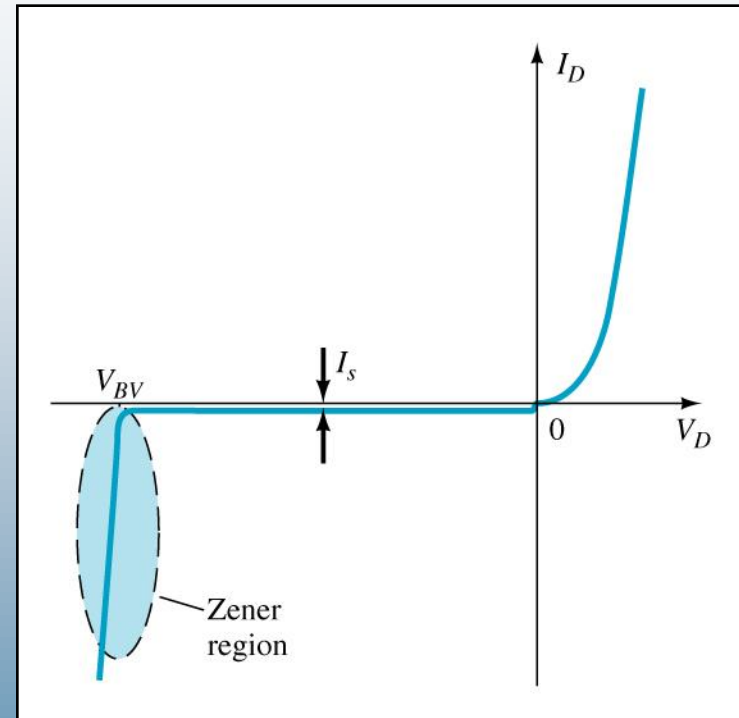
The minority carriers in **n-type** materials are holes.

The minority carriers in **p-type** materials are electrons.

# Breakdown Region

The breakdown region is in the diode's reverse-bias region.

- At some point the reverse bias voltage is so large (at breakdown voltage  $V_{BV}$ ) the diode breaks down and the reverse current increases dramatically.
- This can be **destructive** if the power dissipated exceeds the "safe" level
- The maximum reverse voltage that won't take a diode into the zener region is called the **peak inverse voltage** or **peak reverse voltage**.
- Example: If breakdown voltage  $V_{BV}$  of a diode is  $-60V$ , the peak inverse voltage or peak reverse voltage will be  $-59V$ .



# Forward Bias Voltage/ Threshold Voltage/ Knee Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the p-n junction. This energy comes from the external voltage applied across the diode.

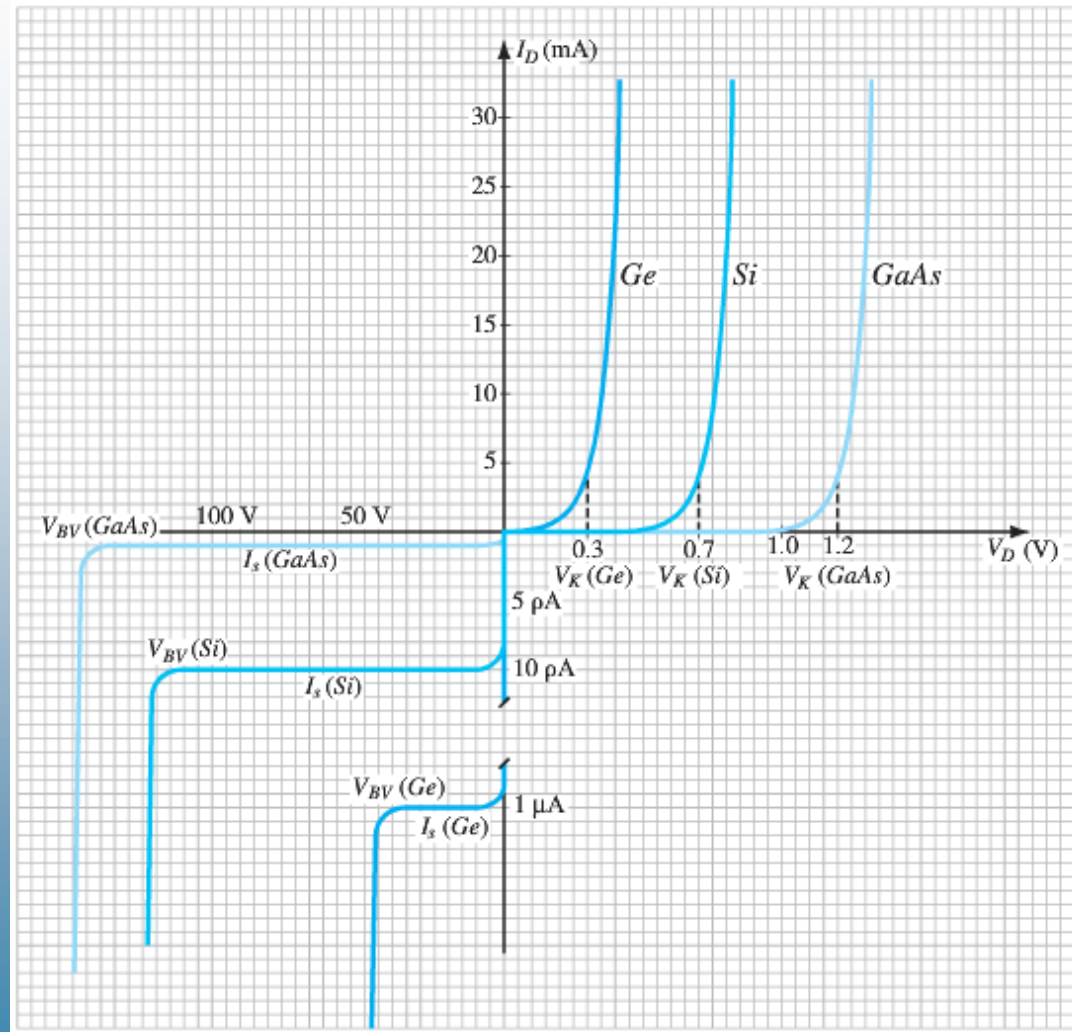
The forward bias voltage required for a:

**gallium arsenide (GaAs) diode  $\cong 1.2$  V**

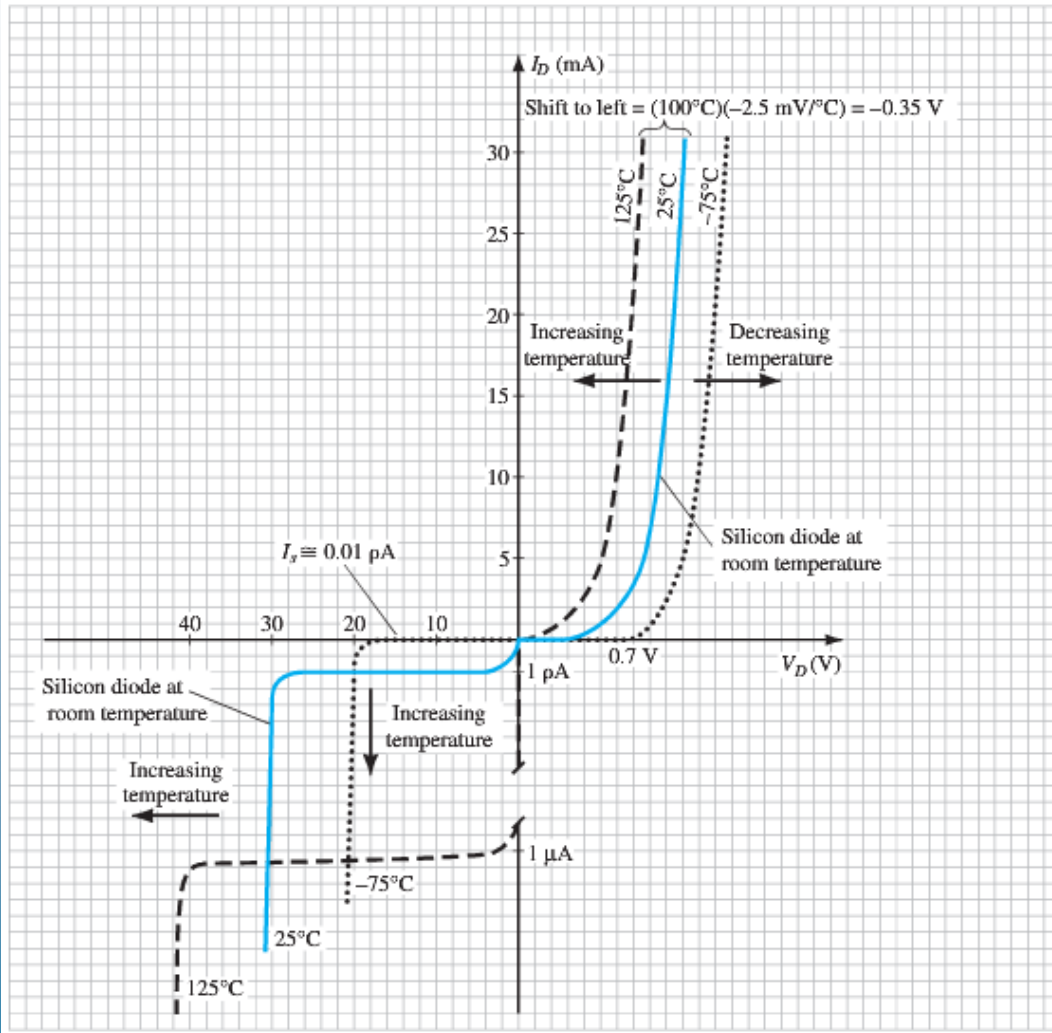
**silicon (Si) diode  $\cong 0.7$  V**

**germanium (Ge) diode  $\cong 0.3$  V**

# Comparison of Ge, Si, and GaAs diodes



# Temperature Effects



- As temperature increases it adds energy to the diode.
  - It reduces the required forward bias voltage
  - It increases the amount of reverse current and reverse breakdown voltage.
- The reverse current of a silicon diode doubles for every  $10^\circ\text{C}$  rise in temp.
- **Germanium diodes are more sensitive to temperature variations than silicon or gallium arsenide diodes.**

# Other Types of Diodes

There are several types of diodes besides the standard  $p$ - $n$  junction diode. Three of the more common are:

**Zener diodes**

**Light-emitting diodes (LED)**

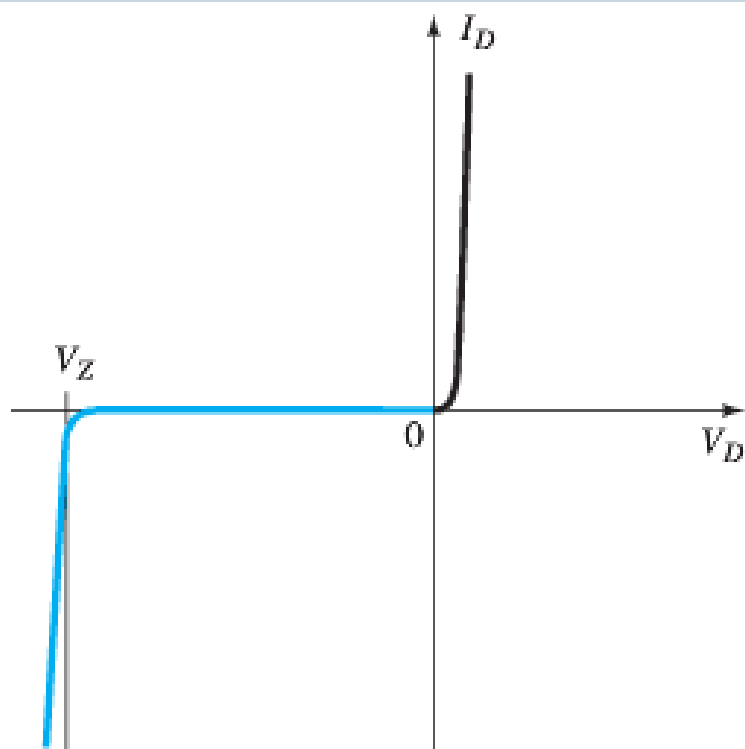
# 1.15 Zener Diode

A **Zener diode** is one that is designed to safely operate in its zener region; i.e., biased at the Zener voltage ( $V_Z$ ).

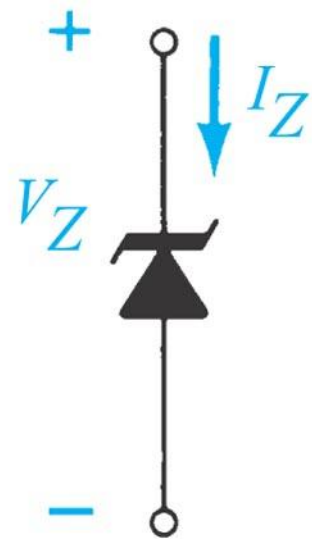
FB region: same as semiconductor diodes.

Breakdown/Zener region: Different from semiconductor diodes.

Common Zener diode voltage ratings are between 1.8V and 200V.

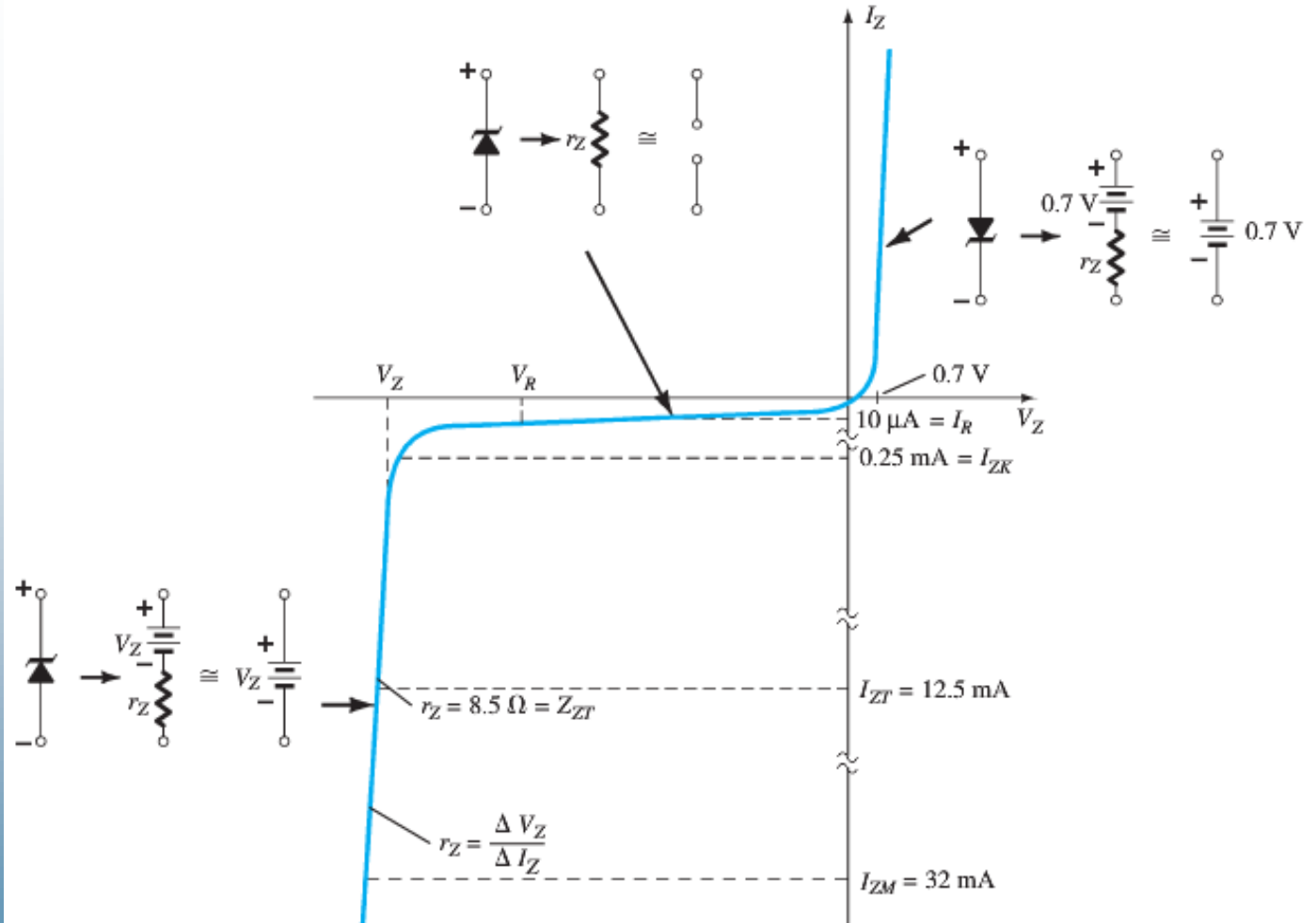


Higher doping than the semiconductor diodes.



# Zener Diode Characteristics

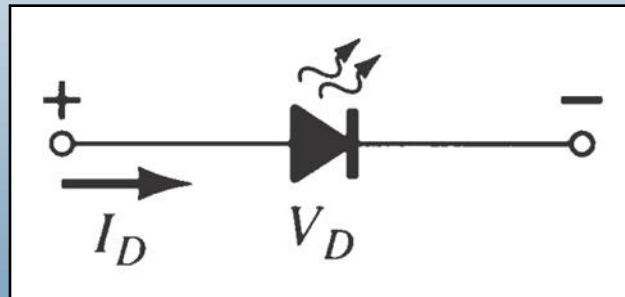
- Typical specifications for a 10-V, 500-mW, 20% Zener diode
- Expected to vary as 10 V + 20%, or from 8 V to 12 V.
- Both 10% and 50% diodes are also readily available.



| Zener<br>Voltage<br>Nominal<br>$V_Z$<br>(V) | Test<br>Current<br>$I_{ZT}$<br>(mA) | Maximum<br>Dynamic<br>Impedance<br>$Z_{ZT}$ at $I_{ZT}$<br>( $\Omega$ ) | Maximum<br>Knee<br>Impedance<br>$Z_{ZK}$ at $I_{ZK}$<br>( $\Omega$ )<br>(mA) | Maximum<br>Reverse<br>Current<br>$I_R$ at $V_R$<br>( $\mu$ A) | Test<br>Voltage<br>$V_R$<br>(V) | Maximum<br>Regulator<br>Current<br>$I_{ZM}$<br>(mA) | Typical<br>Temperature<br>Coefficient<br>(%/°C) |
|---------------------------------------------|-------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------|-----------------------------------------------------|-------------------------------------------------|
| 10                                          | 12.5                                | 8.5                                                                     | 700<br>0.25                                                                  | 10                                                            | 7.2                             | 32                                                  | +0.072                                          |

# 1.16 Light-Emitting Diode (LED)

- An **LED** emits light when it is forward biased, which can be in the infrared or visible spectrum.
- LED and semiconductor diodes work exactly similar in reverse bias region.



The forward bias voltage is usually in the range of 1.5 V to 2.5 V.

# Light-Emitting Diodes

| Color  | Construction | Typical Forward Voltage (V) |
|--------|--------------|-----------------------------|
| Amber  | AlInGaP      | 2.1                         |
| Blue   | GaN          | 5.0                         |
| Green  | GaP          | 2.2                         |
| Orange | GaAsP        | 2.0                         |
| Red    | GaAsP        | 1.8                         |
| White  | GaN          | 4.1                         |
| Yellow | AlInGaP      | 2.1                         |

# Causes of Light in LEDs

- In a forward-biased LED, free electrons cross the junction and fall into holes
  - As these electrons fall from a higher to a lower energy level, they radiate energy
  - In an LED, the energy is radiated as light
  - The color of the light depends on the bandgap of the diode material

# 1.8 Resistance Levels

Semiconductors react differently to DC and AC currents.

*There are three types of resistance:*

**DC (static) resistance**

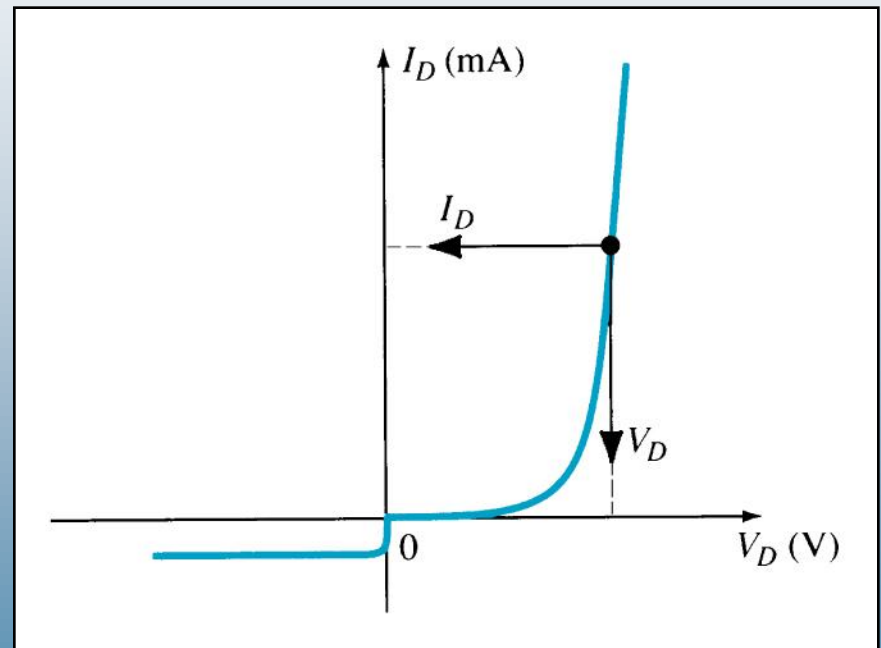
**AC (dynamic) resistance**

**Average AC resistance**

# DC (Static) Resistance

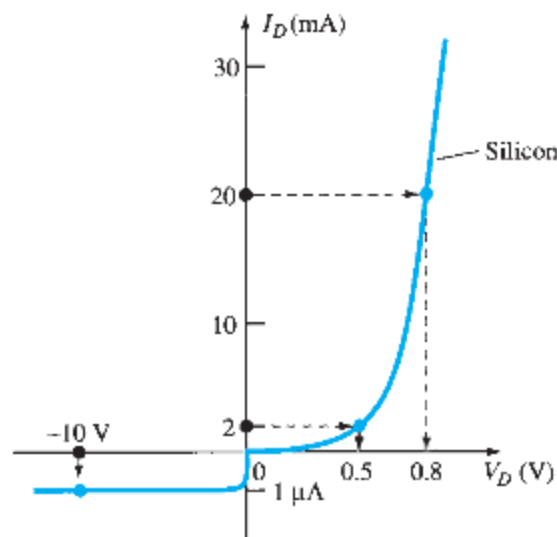
For a specific applied DC voltage ( $V_D$ ) the diode has a specific current ( $I_D$ ) and a specific resistance ( $R_D$ ).

$$R_D = \frac{V_D}{I_D}$$



**EXAMPLE 1.3** Determine the dc resistance levels for the diode of Fig. 1.24 at

- $I_D = 2 \text{ mA}$  (low level)
- $I_D = 20 \text{ mA}$  (high level)
- $V_D = -10 \text{ V}$  (reverse-biased)



**FIG. 1.24**

Example 1.3.

**Solution:**

- a. At  $I_D = 2 \text{ mA}$ ,  $V_D = 0.5 \text{ V}$  (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.5 \text{ V}}{2 \text{ mA}} = \mathbf{250 \, \Omega}$$

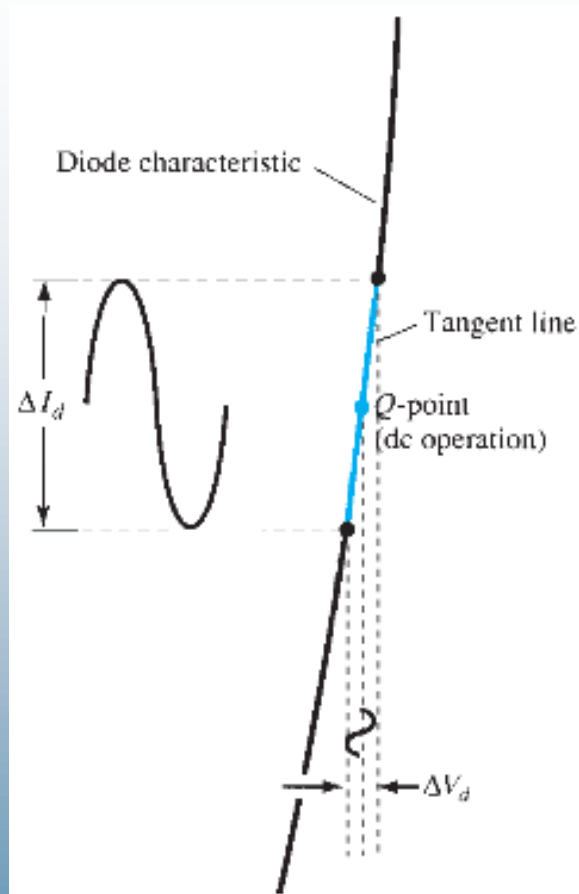
- b. At  $I_D = 20 \text{ mA}$ ,  $V_D = 0.8 \text{ V}$  (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.8 \text{ V}}{20 \text{ mA}} = \mathbf{40 \, \Omega}$$

- c. At  $V_D = -10 \text{ V}$ ,  $I_D = -I_s = -1 \mu\text{A}$  (from the curve) and

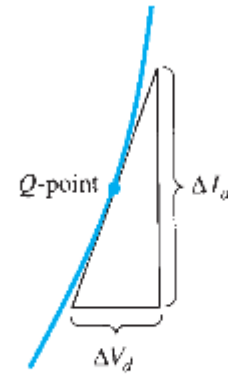
$$R_D = \frac{V_D}{I_D} = \frac{10 \text{ V}}{1 \mu\text{A}} = \mathbf{10 \text{ M}\Omega}$$

# AC (Dynamic) Resistance



**FIG. 1.25**

*Defining the dynamic or ac resistance.*



**FIG. 1.26**

*Determining the ac resistance at a Q-point.*

$$r_d = \frac{\Delta V_d}{\Delta I_d}$$

In the forward bias region:  $r_d = \frac{nV_T}{I_D}$

The resistance  $r_d$  depends on the amount of current ( $I_D$ ) in the diode.

# Analytical Derivation of $r_d$

$$\frac{d}{dV_D}(I_D) = \frac{d}{dV_D}[I_S(e^{V_D/nV_T} - 1)]$$

$$\frac{dI_D}{dV_D} \cong \frac{I_D}{nV_T}$$

Flipping the result to define a resistance ratio ( $R = V/I$ ) gives

$$\frac{dV_D}{dI_D} = r_d = \frac{nV_T}{I_D}$$

Substituting  $n = 1$  and  $V_T \cong 26 \text{ mV}$  from Example 1.1 results in

$$r_d = \frac{26 \text{ mV}}{I_D}$$

*the dynamic resistance can be found simply by substituting the quiescent value of the diode current into the equation.*

# Total AC (Dynamic) Resistance

The total ac resistance  $r'_d$  includes  $r_B$ , which is the resistance of the semiconductor material itself (called *body resistance*) and the resistance introduced by the connection between the semiconductor material and the external metallic conductor (called *contact resistance*).

$$r'_d = r_d + r_B = \frac{V_T}{I_D} + r_B = \frac{26 \text{ mV}}{I_D} + r_B$$

$r_B$  ranges from a typical  $0.1 \Omega$  for high power devices to  $2 \Omega$  for low power, general purpose diodes. In some cases  $r_B$  can be ignored.

In the reverse bias region:

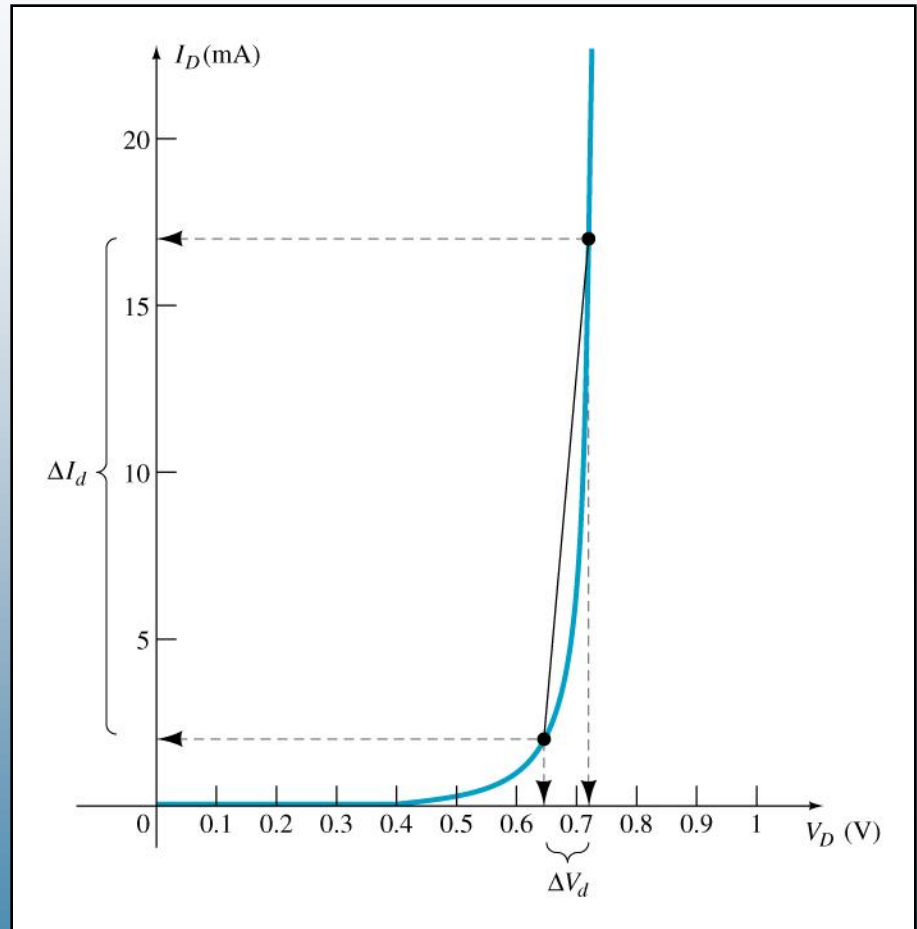
$$r'_d = \infty$$

***The resistance is effectively infinite. The diode acts like an open.***

# Average AC Resistance

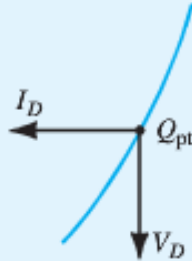
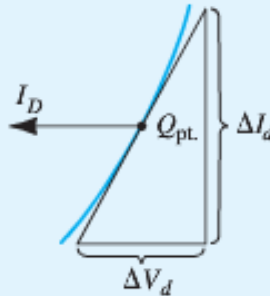
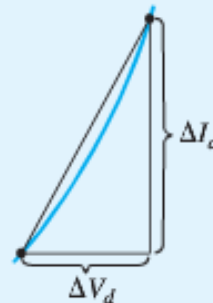
$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \quad | \quad pt. to pt.$$

AC resistance can be calculated using the current and voltage values for two points on the diode characteristic curve.



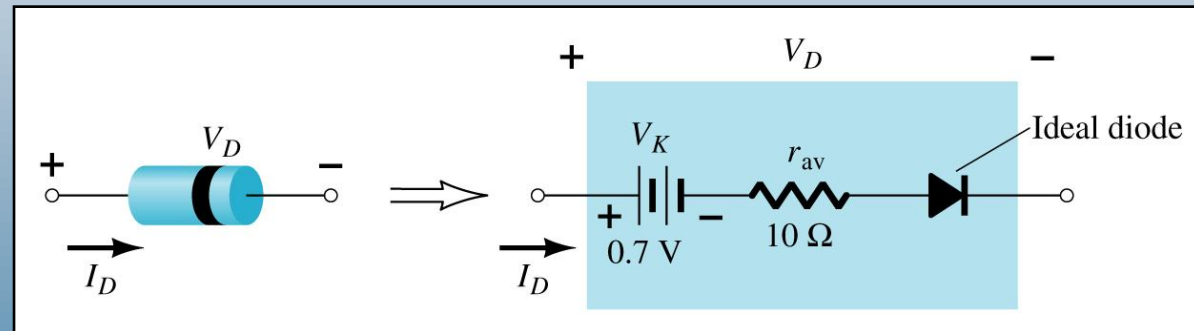
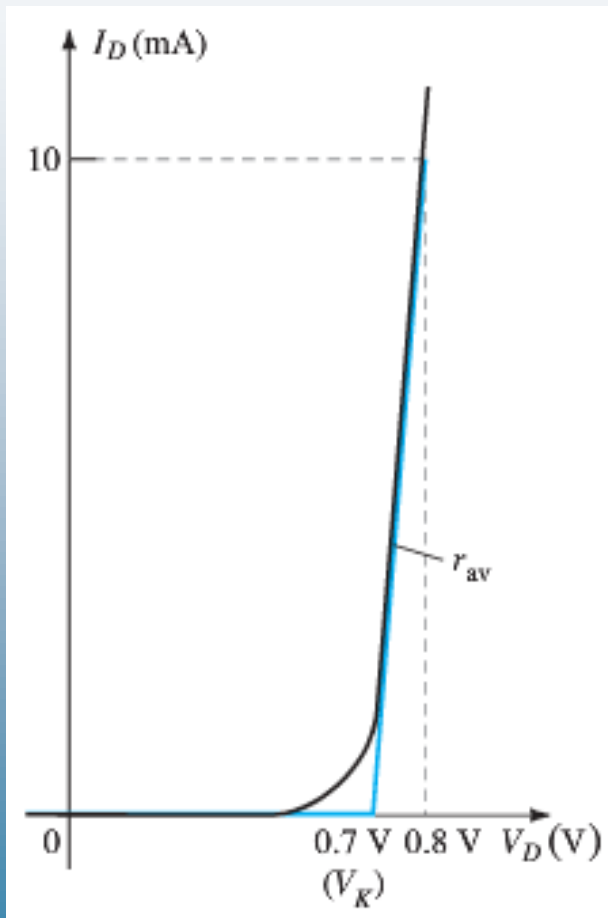
# Summary Table

*Resistance Levels*

| Type          | Equation                                                                    | Special Characteristics                                | Graphical Determination                                                              |
|---------------|-----------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|
| DC or static  | $R_D = \frac{V_D}{I_D}$                                                     | Defined as a point on the characteristics              |   |
| AC or dynamic | $r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26 \text{ mV}}{I_D}$           | Defined by a tangent line at the $Q$ -point            |   |
| Average ac    | $r_{av} = \left. \frac{\Delta V_d}{\Delta I_d} \right _{\text{pt. to pt.}}$ | Defined by a straight line between limits of operation |  |

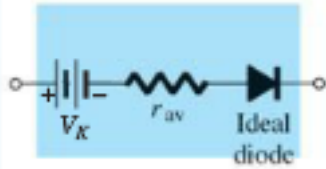
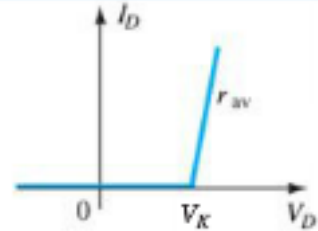
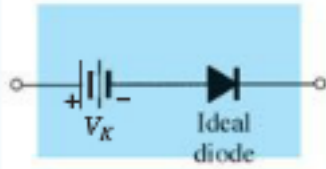
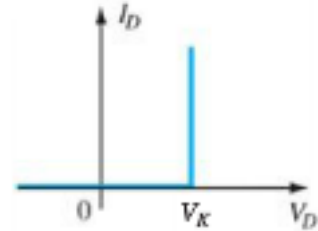
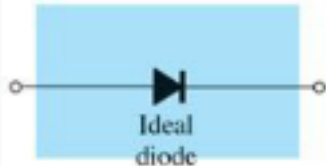
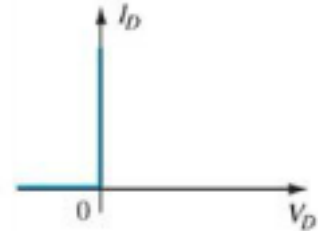
# 1.9 Diode Equivalent Circuit

- Combination of **properly chosen elements** to represent the actual characteristics of a device.



# Summary Table

## Diode Equivalent Circuits (Models)

| Type                                                    | Conditions                                                             | Model                                                                                | Characteristics                                                                      |
|---------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Piecewise-linear model<br>3 <sup>rd</sup> Approximation |                                                                        |    |   |
| Simplified model<br>2 <sup>nd</sup> Approximation       | $R_{\text{network}} \gg r_{\text{av}}$                                 |    |   |
| Ideal device<br>1 <sup>st</sup> Approximation           | $R_{\text{network}} \gg r_{\text{av}}$<br>$E_{\text{network}} \gg V_K$ |  |  |

# Practice Problem

- Use 11<sup>th</sup> edition of the textbook
- Total points: 100 (10 each)
- Problems:
  - ***Chapter 1***: 15, 16, 18, 19, 27, 28, 29, 31

# Quiz 1

- Syllabus:
  - Chapter 1

# Quiz 1

- Quiz 1 11/09/2020 Wednesday ??
- Syllabus: Chapter 1 Semiconductor Diodes
- There will be 1 math problem + Short questions/MCQs/True False.
- Math Problem from Diode Current Equation (Slides: 30,31 & 34) OR AC Resistance (Slide 50).

1. Draw V-I characteristics curve of an Ideal Diode or Semiconductor (Silicon/ Germanium) Diode.
2. What do you mean by intrinsic semiconductors?
3. What do you mean by extrinsic semiconductors?
4. What are the majority and minority carriers of p-type and n-type materials?
5. What happens in a semiconductor diode in no bias condition?
6. What happens in a semiconductor diode in the forward bias condition?
7. What is break down/ Zener voltage? What is peak inverse voltage (PIV)?
8. What is forward bias voltage/knee voltage?
9. Draw the symbol of a semiconductor diode, Zener diode, and LED.
10. What are the differences between semiconductor (Si) diode and Zener diode?